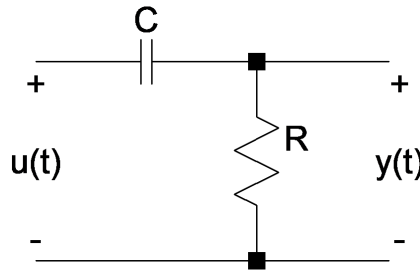


ELE228 - Programming for Circuits and Systems Preliminary Work 4**1 Questions**

1. Consider the circuit in *Fig. 1*, derive the differential equation describing the input-output characteristics, where $u(t)$ is the input and $y(t)$ is the output.
2. Compute the impulse response of the system. Implement the impulse response $h(t)$ in MATLAB and plot it against time.
3. Implement the convolution sum in MATLAB. Create an impulse signal and convolve it with the impulse response $h(t)$. Compare the output with the impulse response signal.

*Figure 1*

$R = 20 \, \Omega$ and $C = 0.5 \cdot 10^{-3} \, \text{F}$.

2 Answers

1. From KVL,

$$\frac{1}{C} \int_0^t i(\tau) d\tau + y(t) = u(t)$$

Differentiate each side with respect to t .

$$\frac{i(t)}{C} + y'(t) = u'(t) \implies \boxed{y'(t) + \frac{y(t)}{RC} = u'(t)}$$

For $R = 20 \, \Omega$ and $C = 0.5 \cdot 10^{-3} \, \text{F}$,

$$\boxed{y'(t) + 100y(t) = u'(t)}$$

2. The transfer function is given by

$$H(s) = \frac{Y(s)}{U(s)},$$

where $Y(s) = \mathcal{L}\{y(t)\}$ and $U(s) = \mathcal{L}\{u(t)\}$. The s -domain equivalent resistance of the capacitor: $\frac{2000}{s}$. From the voltage divider,

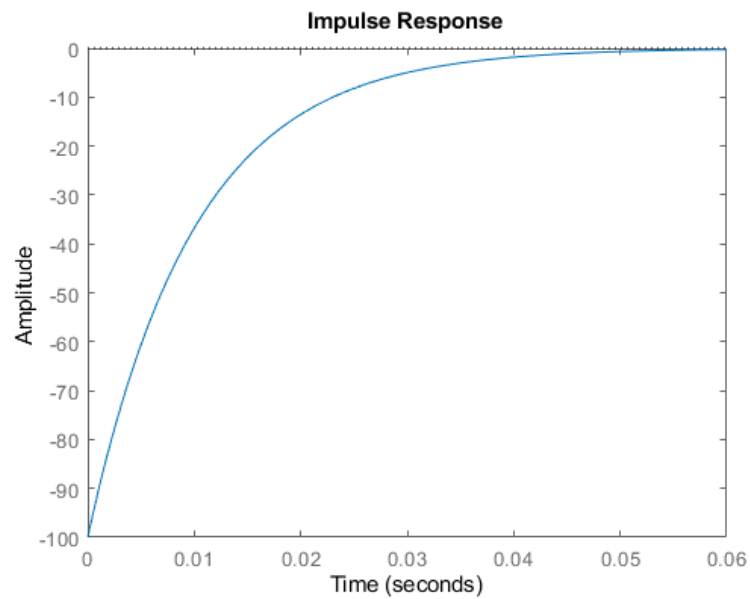
$$Y = U \cdot \frac{20}{20 + \frac{2000}{s}} \implies H = \frac{Y}{U} = \frac{s}{s + 100} = 1 - \frac{100}{s + 100}$$

$$\implies h(t) = \mathcal{L}^{-1}\{H(s)\} = \delta(t) - 100e^{-100t}u(t)$$

Code:

```
num = [1 0];
denom = [1 100];

H_tf = tf(num, denom);
figure
impulse(H_tf)
```



3. Code:

```
hold on
syms s

H_sym = poly2sym(num,s) / poly2sym(denom,s);
h = ilaplace(H_sym)

t = 0:10e-4:0.06;

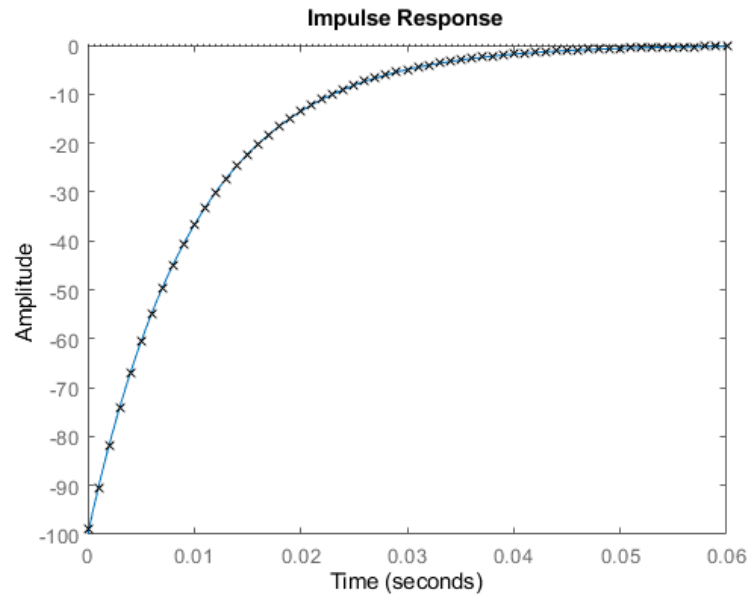
delta_t = zeros(size(t));
delta_t(1) = 1;

h_t = delta_t - 100*exp(-100*t);
h_conv = conv(delta_t, h_t);

plot(t, h_conv(1:length(t)), 'xk')
```

h =

dirac(t) - 100*exp(-100*t)



The graphs are almost identical. However, because the Dirac delta vector is discrete in MATLAB, the value of the first sample that is convolved is -99 . Anything else is as anticipated and hence confirms the previous result.